

Nestor David Rendon, PhD

| Data Science | Ecoacoustics | Biodiversity & Landscape Analysis

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Professional Profile

My research is situated at the interface of **soundscape ecology** and quantitative environmental science. Conceptually, I treat soundscapes as **multivariate acoustic systems** in which temporal covariation among spectral and temporal descriptors carries information about landscape-scale ecological structure and its dynamics; integrating such evidence with other sensors and covariates can tighten inference where labels are incomplete. I also work on the **transdisciplinary implications** of acoustic monitoring: how rigorously derived soundscape indicators can enter deliberation with land managers, conservation actors, and public-health-oriented colleagues, so that biophysical change and human experience are not discussed in isolation.

I hold a PhD in Electrical & Computer Engineering (specialisation: Data Science and Statistical Modelling) centred on **passive acoustic monitoring** and the construction of **interpretable, landscape-level indicators** from large audio and environmental datasets under weak supervision. Representative publications include an unsupervised framework for mapping **landscape acoustic heterogeneity** (*Environmental Modelling & Software*, 2025) and complementary methods to quantify **acoustic heterogeneity across ecological soundscapes** (*Ecological Indicators*, 2022). In parallel, at **xFarm** (Switzerland) I develop production-grade **Python/R** workflows, computer-vision and deep-learning components, and operational monitoring products including **insect monitoring** and satellite- or sensor-based diagnostics that must remain robust under real-world heterogeneity and scale. The domain differs from forest-field campaigns, but the inferential problem is analogous: distilling high-dimensional evidence into decision-relevant signals.

Career trajectory. Following this industry phase, I intend to return to **academic research** and to concentrate my effort on **ecosystem transformation**, biodiversity-relevant monitoring, and soundscape-oriented questions at the science–policy–society interface. I aim to combine peer-reviewed advances in ecoacoustics with the software-engineering discipline acquired in industry.

Professional Experience

xFarm Technologies

Switzerland

Vision engineer – Data Scientist

2020–Present

- Design and implementation of conversational agents and deep learning components for large-scale agricultural and environmental monitoring.
- End-to-end ML pipelines: ingestion, preprocessing, training, validation, deployment; monitoring of inference in cloud (AWS, GCP).
- Modular integration of vision and ML models into operational systems; experimentation with datasets for multi-agent workflows.
- Containerization, version control, and structured development practices.

University of Antioquia

Colombia

PhD Researcher – Machine Learning & Statistical Modeling

2021–2024

- Unsupervised learning methodologies for **soundscape analysis** and landscape-level acoustic patterns.
- Novel internal clustering validation (Fréchet distance); applications to ecological and high-dimensional data.
- Modular frameworks for large-scale analysis of geospatial, audio, and time-series environmental data.
- Statistical modelling of high-dimensional datasets including **passive acoustic monitoring**.
- Peer-reviewed publications in ecological modelling, applied AI, and ecological indicators (see Selected Publications).

Laboratoire MIA

Machine Learning Engineer

France
2022–2023

- Benchmarking frameworks for unsupervised learning algorithms.
- Experimental evaluation and modular ML pipelines, including ecoacoustic-oriented analyses.
- Collaboration on interdisciplinary AI research projects.

SUPSI - DTI (Reteca foundation internship)

Machine Learning Engineer

Switzerland
2020–2021

- Deep learning pipelines for object detection; software modules for data acquisition and preprocessing.
- Collaboration with industrial partners on applied ML solutions.

IES and Colombia de Colores corporation

Software / ML engineer and teacher

Colombia
2017–2020

- Software development for regulated systems; ML modules for rain detection in acoustic recordings.
- Programming instruction.

Education

University of Antioquia

PhD in Electrical and Computer Engineering: Data Science & Statistical Modeling (Cum Laude)

2024

University of Antioquia

MSc in Machine Learning

2020

University of Antioquia

BSc in Electronic Engineering

2017

Selected Publications

Rendon et al. (2025). *Letting Ecosystems Speak for Themselves: An Unsupervised Methodology for Mapping Landscape Acoustic Heterogeneity*. Environmental Modelling & Software

Rendon et al. (2023). *Uncertainty Clustering Internal Validity Assessment using Fréchet Distance*. Engineering Applications of Artificial Intelligence.

Rendon et al. (2022). *Automatic Acoustic Heterogeneity Identification*. Ecological Indicators.

PhD thesis (URL on Google Drive)

Google Scholar profile

Technical Skills

Programming: Python (advanced), R, C++, SQL, Bash

Ecoacoustics / data: Passive acoustic monitoring, time-series and spatial audio analysis, clustering and validation

AI & ML: Deep learning, unsupervised learning, statistical modelling, representation of high-dimensional data

Systems: Modular pipelines, MLOps-oriented workflows, Docker, Git, CI/CD

Frameworks: QGIS, PyTorch, TensorFlow, Scikit-learn, OpenCV, FastAPI; Agent Development Kit (ADK)

Cloud & tools: AWS, GCP, Grafana, Metabase, Power BI, CUDA, documentation (Sphinx)

Research: Experimental design, benchmarking, reproducible analysis

Languages

Spanish (native)
Italian (C1)
English (B2; scientific communication)
French (A2)

References

Prof. Abel Paz

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Prof. Claudia Victoria Isaza

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